

Answer all questions.

Figures at the right margin indicate marks. All parts of a question must be answered at one place.

A mere answer without showing intermediate steps will not be credited.

1. (a) Build a linear regression model using LMS algorithm to find a best fit to the following equation. CO1 BL3 [5]

$$Y = 2 + 3x_1 + 2x_2$$

Use at least four data samples to find the regression parameters. Perform two iterations of Batch Gradient Descent to update the parameters. Also compute MSE after 2 iterations.

- (b) Prove that sum $z = x + y$ of two independent random variables x , and y where $x \sim N(\mu_x, \sigma_x^2)$ and $y \sim N(\mu_y, \sigma_y^2)$ is also Gaussian with mean value and variance equal to $\mu_x + \mu_y$, and $\sigma_x^2 + \sigma_y^2$. CO2 BL3 [3]
- (c) Find the discriminant function for equal covariance matrix assuming posterior probability as the discriminant. CO1 BL2 [2]
2. (a) Consider a 2-class classification task in the 2-dimensional space, where the classes ω_1 and ω_2 are distributed according to CO3 BL3 [5]

$$p(x|\omega_1) = \frac{1}{(\sqrt{2\pi\sigma_1^2})^2} \exp\left(-\frac{1}{2\sigma_1^2} (x - \mu_1)^T (x - \mu_1)\right), p(x|\omega_2) = \frac{1}{(\sqrt{2\pi\sigma_2^2})^2} \exp\left(-\frac{1}{2\sigma_2^2} (x - \mu_2)^T (x - \mu_2)\right)$$

$\mu_1 = [1, 1]^T, \mu_2 = [1.5, 1.5]^T, \sigma_1^2 = \sigma_2^2 = 0.2$. Assuming the classes are equiprobable, design a Bayesian classifier that (i) minimizes the average risk with loss matrix $L =$

$$\begin{bmatrix} 0 & 1 \\ 0.5 & 0 \end{bmatrix}. \text{ (ii) Minimizes the error probability.}$$

- (b) Use a Naive Bayes classifier to determine whether or not someone with excellent attendance, poor GPA, and lots of effort should be hired. CO4 BL3 [3]

Name	GPA	Effort	Hirable
Sarah	poor	lots	Yes
Dana	average	some	No
Alex	average	some	No
Annie	average	lots	Yes
Emily	excellent	lots	Yes
Pete	excellent	lots	No
John	excellent	lots	No
Kathy	poor	some	No

- (c) Define R-squared and Adjusted R-squared used in regression. CO2 BL2 [2]
3. (a) Below you are given a summary of the output from a simple linear regression analysis from a sample of size 15, SSR=100, SST = 152. Find the coefficient of determination. CO1 BL3 [2]
- (b) How do you validate a MLR model? Discuss. CO2 BL1 [2]
- (c) Explain logistic regression with an example. CO2 BL2 [2]
- (d) What is a confusion matrix? How ROC of a classifier is plotted? CO1 BL3 [2]
- (e) What is gradient Descent? Write the Gradient Descent rule for minimizing a function. CO2 BL1 [2]